SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.1: Introduction to General Science — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 1.1: Introduction to General Science
Driving Question	DRIVING QUESTION: How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **How Did She Know? — Anchoring the Phenomenon & Opening the Driving Question Board**	Students observed the anchoring phenomenon: a Kenyan farmer (e.g., Mama Akinyi from Kisumu) accurately diagnoses poor soil fertility and predicts a weak maize harvest — without a laboratory, without seeing the soil chemistry directly. Teacher presents photographic or video evidence of yellowing maize stalks, stunted growth patterns, and cracked topsoil alongside the farmer's confident diagnosis. Students also observe a short parallel case: a Kenyan doctor in a rural Kericho clinic inferring malaria from visible symptoms before a blood test confirms it. Pedagogical note: Ensure ALL students see and interact with the stimulus before any discussion begins — display images prominently and allow 2 minutes of silent individual observation first.	Students learned that experts — scientists, farmers, and doctors alike — routinely reach reliable conclusions about things they cannot see directly (soil chemistry, pathogens, atomic structures) by reading visible patterns and evidence. They learned that this inferential power is the foundation of science and is practiced daily in Kenya. Students also generated their first personal questions for the Driving Question Board (DQB), revealing prior knowledge gaps the teacher can use diagnostically. Pedagogical note: The DQB questions students generate here are DATA — photograph or record them. They will be revisited and refined in Lessons 7 and 8 to demonstrate cumulative learning.	The phenomenon is explained by the concept of INFERENCE: a logical conclusion drawn from observable evidence plus prior knowledge. Mama Akinyi's diagnosis is not a guess — it is inference. Yellow lower leaves + stunted growth + sandy, pale topsoil are visible indicators that map predictably onto nitrogen-deficient soil chemistry she cannot see. The doctor's inference works the same way. Teacher should cold-call at least 4 students to articulate in their own words: 'What did she SEE, and what did she CONCLUDE?' Enforce the observe/infer distinction explicitly here — it anchors every lesson that follows. Pedagogical note: Write 'OBSERVE' and 'INFER' as permanent anchor words on the board for the entire sub-strand.
Lesson 2 **What is General Science? Mapping the Discipline**	Students observed a structured 'science map' stimulus: a large visual showing three branching disciplines — Biology, Chemistry, and Physics — each illustrated with Kenyan-relevant examples (Biology: Lake Victoria fish species, malaria parasite;	Students learned that General Science is a systematic, evidence-based discipline for asking and answering questions about the natural world — not merely a collection of memorised facts. They learned the three major sub-disciplines (Biology, Chemistry,	The Kisumu maize case is explained by showing how a complete scientific investigation of poor yield draws on Biology (plant cell nutrient uptake), Chemistry (soil nitrogen compounds, pH), and Physics (water drainage, sunlight energy). No single

	Chemistry: fertiliser used in Rift Valley farms, the chemistry of ugali starch; Physics: M-Pesa mobile signal transmission, Nairobi solar panels). Students also observed a short case study of the Kisumu maize investigation — noting how a single real-world problem (poor harvest) draws on all three disciplines simultaneously. Pedagogical note: Give students 3 minutes to annotate the map individually before paired discussion. This activates prior knowledge and surfaces misconceptions early.	Physics) and identified at least two Kenyan real-world examples within each. They also learned the key distinction: science describes and explains HOW and WHY natural phenomena occur, while technology applies that knowledge. Pedagogical note: Students frequently conflate science with technology — address this directly using the maize example: understanding why nitrogen is deficient (science) vs. designing a slow-release fertiliser pellet (technology).	discipline is sufficient alone — General Science is integrative. Students explain back: 'Which discipline answered which part of the maize problem?' Teacher waits a minimum of 5 seconds after each question before accepting answers (structured wait time). Cold-call at least 3 students who have not yet volunteered. Pedagogical note: This lesson plants the 'integration' idea that becomes critical in Lesson 3 when students examine real Kenyan career paths that cross disciplinary boundaries.
Lesson 3 **Importance of General Science in Life, Environment and Technology**	Students observed three curated evidence sets — each representing one domain of Kenyan life: (1) HUMAN LIFE: data cards showing Kenya's malaria reduction statistics post-2000 (attributable to scientific intervention — bed nets, artemisinin treatment); iron-deficiency anaemia rates linked to diet (ugali + sukuma wiki nutritional data). (2) ENVIRONMENT: before/after images of Karura Forest restoration, Lake Victoria water hyacinth invasion data, and a graph of Nairobi air quality trends. (3) TECHNOLOGY: images of M-Pesa infrastructure, JKUAT-developed drought-resistant maize varieties, and Kenya's geothermal energy plant at Olkaria. Pedagogical note: Rotate evidence sets among groups so all three domains are processed by the whole class — use a jigsaw structure for efficiency.	Students learned that General Science operates across three interconnected domains in Kenya: human life (nutrition, disease prevention, medicine), environment (conservation, pollution monitoring, climate adaptation), and technology (agricultural innovation, energy, communications). They learned that scientific knowledge is not neutral — it has direct, measurable consequences for Kenyan communities. Students also learned that Kenya's constitutional obligation to a clean environment is supported by scientific monitoring and evidence. Pedagogical note: Students should be able to give at least one specific Kenyan example per domain from memory by the end of this lesson — this is a useful exit-ticket check.	The three domains are explained through causal chains: Science DISCOVERS a mechanism → Technology APPLIES it → Life and Environment are CHANGED. Example chain: Research identifies malaria parasite lifecycle (Biology) → Chemistry synthesises artemisinin → Kenyan children's survival rates improve. A second chain: Physics and Chemistry identify greenhouse gas absorption → Engineers design Olkaria geothermal plant → Kenya reduces fossil fuel dependence. Teacher explicitly asks: 'How does this connect back to Mama Akinyi's farm?' (Answer: her soil problem sits at the intersection of all three domains.) Pedagogical note: Use this lesson to challenge the misconception that science is only done in laboratories by people in white coats — Kenyan evidence powerfully counters this.
Lesson 4 **Careers in General Science: Research, Chart Making & Career-Path Justification**	Students observed four short career-profile cards of real or representative Kenyan science professionals: (1) Dr. Daisy Rutto — plant biologist, Kenya Agricultural and Livestock Research Organisation (KALRO), develops drought-tolerant crops. (2) A clinical chemist at Kenyatta National Hospital interpreting blood chemistry panels. (3) A Kenya Meteorological Department physicist using atmospheric data to forecast Rift Valley rainfall. (4) An environmental scientist monitoring Lake Victoria pollution levels. Each card shows:	Students learned that a wide range of well-defined, valued Kenyan careers are built on foundations of Biology, Chemistry, and Physics — including medicine, agricultural research, environmental monitoring, meteorology, food science, engineering, and science education itself. They learned to map specific science sub-topics to specific career pathways and to justify that mapping with evidence. Students also produced a personal career-path chart connecting their own interests to General Science sub-topics. Pedagogical note: The	Career paths are explained through the lens of the sub-strand's driving question: each career listed is a professional form of inference — the meteorologist infers tomorrow's weather from today's atmospheric data; the clinical chemist infers disease from blood chemistry the eye cannot see. Teacher connects explicitly: 'Every career on your chart is a person whose job is to answer questions using evidence — exactly what we have been practising.' Cold-call 5 students: 'Name your chosen career and name ONE thing that

	job title, daily tasks, science sub-disciplines used, and a Kenyan institution employing that role. Pedagogical note: Allow students to read cards silently first, then discuss in pairs — avoid reading aloud by the teacher, which reduces student engagement.	3-sentence career justification students write is a formative writing-in-science task — it should use the word 'evidence' and name at least one specific science concept. Collect and give written feedback.	professional cannot see directly but must infer.' Pedagogical note: This lesson has strong PCIs (career guidance, life skills) — acknowledge explicitly that science careers in Kenya are accessible and increasingly well-compensated.
Lesson 5 **Introduction to Scientific Processes and the Principle of Inference (Explain Phase – Part 1)**	Students observed two carefully designed evidence sets that force inference without direct observation: (1) A sealed 'mystery envelope' containing only indirect clues — a photograph of footprints in Tsavo East sand, droppings, broken acacia branches, and a partial shadow — from which students must infer the animal. (2) A chemistry scenario: a Kenyan chemist receives a clear, odourless liquid sample from a Mombasa water source — students are given only pH data, conductivity readings, and a boiling-point measurement and must infer whether it is safe drinking water. Pedagogical note: Do NOT reveal answers immediately — sustained uncertainty is pedagogically productive here. Students need to sit with incomplete information before the explanation phase.	Students learned the formal definition of inference: a conclusion drawn from available evidence combined with prior knowledge and logical reasoning — explicitly NOT a direct observation and NOT a random guess. They learned the scientific process steps (observation → question → hypothesis → experiment → data collection → analysis → inference/conclusion) and identified WHERE inference sits in that process. Students also learned the difference between a STRONG inference (multiple converging evidence lines) and a WEAK inference (single piece of ambiguous evidence). Pedagogical note: The observe/infer distinction introduced in Lesson 1 is now formalised — return to Mama Akinyi's diagnosis and ask students to re-label each step using the formal process vocabulary.	Inference is explained using the concept of CONVERGENT EVIDENCE: the more independent lines of evidence that point to the same conclusion, the stronger the inference. In the Tsavo footprint case: large circular pad + heavy branch breakage + acacia bark damage + size of shadow → elephant (not rhino, not buffalo). In the water chemistry case: pH 7.1 + low conductivity + 100°C boiling point → consistent with pure water (but teacher notes: inference is not certainty — a field test would still be required). Teacher explicitly states the epistemological principle: 'Scientists say they KNOW something indirectly when multiple independent evidence sources converge on one explanation.' Cold-call 4 students to apply this principle to a new scenario.
Lesson 6 **Collecting Evidence and Drawing Inferences: Mystery Bags Hands-On Activity**	Students conducted the Mystery Bags hands-on investigation: sealed paper bags containing common Kenyan household/farm objects (examples: a small dried maize cob, a piece of sisal rope, a metal bottle cap, a smooth river stone from Tana River). Without opening the bag, students systematically used four evidence-collection methods — (1) observation (visual inspection of bag shape/movement), (2) measurement (mass using a school balance, dimensions by feel), (3) indirect experimentation (tilting, shaking, pressing), (4) record-keeping (structured data table completed in real time). Each group recorded observations in a formal data table before making any inference. Pedagogical note: Enforce the rule strictly: NO INFERENCES until the data table is complete. This builds the habit of	Students learned and named the four primary methods scientists use to collect evidence: systematic observation, measurement, controlled experimentation, and rigorous record-keeping. They learned that each method has specific strengths and limitations, and that combining methods produces stronger evidence. They also practised constructing a formal inference statement in the structure: 'Based on [evidence], I infer [conclusion] because [reasoning].' Pedagogical note: The structured inference statement format should be posted permanently in the classroom and required in all subsequent written work in this sub-strand. Assess whether students distinguish between 'I think it is...' (opinion) and 'Based on... I infer...' (evidence-based conclusion).	The mystery bag activity is explained by connecting each evidence-collection method to a real Kenyan scientific context: systematic observation (Kenya Wildlife Service rangers counting wildlife without disturbing herds); measurement (KEMRI researchers measuring parasite loads in blood samples); experimentation (KALRO agronomists testing fertiliser rates on Rift Valley trial plots); record-keeping (Kenya Meteorological Department maintaining 50-year rainfall records that reveal climate change trends). Teacher asks: 'Which of these four methods did Mama Akinyi use on her farm in Lesson 1?' (Answer: observation and prior-knowledge-based inference.) Pedagogical note: This lesson is the practical, kinaesthetic anchor of the sub-strand — students who struggled with abstract inference in Lesson 5 often

	separating data collection from interpretation.		achieve understanding here.
Lesson 7 **DQB Curation and Cumulative Model Building**	Students physically returned to the Driving Question Board questions generated in Lesson 1. Each question was printed or written on a card. Students observed the full set of questions as a class display and were given colour-coded stickers: GREEN = 'We can now answer this with evidence from our lessons'; YELLOW = 'We have partial evidence — we can make a strong inference but cannot fully confirm'; RED = 'This question is still genuinely open — we need more investigation.' Students also observed their own Lesson 1 cumulative models (initial diagrams of 'how a scientist knows something they cannot see') and compared them to their current understanding. Pedagogical note: This comparison of early vs. current models is a powerful metacognitive moment — name it explicitly as 'model revision' and connect it to how real scientists update models when new evidence arrives.	Students learned that scientific understanding is cumulative and necessarily incomplete — answered questions generate new, deeper questions, and this is a feature of science, not a failure. They learned to distinguish between questions that are now answerable with evidence, questions that support a strong inference, and questions that remain genuinely open. Students revised their cumulative explanatory models to incorporate formal vocabulary (inference, convergent evidence, scientific process, observation vs. conclusion) and Kenyan examples gathered across all lessons. Pedagogical note: Model revision is a core NGSS Science and Engineering Practice — frame it explicitly: 'Scientists never throw away a model; they revise it when evidence demands it. That is what you are doing right now.'	The DQB curation process is explained through the sub-strand's central epistemological answer: a scientist knows something is true when multiple independent evidence sources converge on the same explanation, the inference is testable, and the conclusion has been peer-reviewed and revised over time. Students articulate this in their own words as a written synthesis statement: 'A scientist knows something they cannot see directly when...' Teacher cold-calls 6 students to read their synthesis statements aloud and facilitates class comparison. Pedagogical note: Collect synthesis statements as a summative formative assessment — they reveal whether students have grasped the epistemological core of the sub-strand or have remained at the surface level of 'scientists use experiments.'
Lesson 8 **Sub-Strand Review, Assessment and Transfer**	Students observed a new, unseen transfer scenario — deliberately chosen to be outside the contexts used in Lessons 1–7: a Nairobi forensic scientist is presented with trace evidence from a crime scene (a fibre, a chemical residue on a surface, a partial shoeprint in dust) and must reconstruct what happened without having witnessed the event. Students also observed a short data set from Kenya's 2023 national nutrition survey showing rising rates of stunting in arid counties, with no single visible cause. Both scenarios require students to apply all sub-strand learning — inference, convergent evidence, scientific process, evidence-collection methods — in a new context. Pedagogical note: Transfer to a new context is the truest test of understanding — resist the urge to scaffold heavily here. Allow productive struggle for at least 5 minutes before offering support.	Students demonstrated consolidated understanding of the full sub-strand by applying the following to the transfer scenario: (1) the formal definition and structure of scientific inference; (2) the four evidence-collection methods; (3) the principle of convergent evidence; (4) the relationship between General Science and Kenyan life, environment, and careers; (5) the iterative, cumulative nature of scientific knowledge. Students also completed a structured self-assessment against the sub-strand learning outcomes, identifying personal strengths and one area for continued development. Pedagogical note: The self-assessment should be filed in each student's portfolio — it builds metacognitive awareness and models the scientific habit of honest, evidence-based self-evaluation.	The transfer scenarios are explained by returning one final time to the sub-strand driving question: 'How does a scientist know something is true when they cannot see it directly?' Students now answer this with precision: by collecting evidence using systematic methods, identifying convergent independent evidence lines, constructing a structured inference, testing and revising that inference, and situating their conclusion within the broader scientific community. The forensic scientist and the nutritional epidemiologist are doing exactly what Mama Akinyi did — and exactly what every scientist in Kenya's research institutions, hospitals, farms, and conservation areas does every day. Pedagogical note: End the lesson by asking students to write one sentence connecting their personal career-path chart (Lesson 4) to the driving question — this closes the sub-strand narrative arc and gives every student a personalised,

			motivating takeaway.